COVID-19 GLOBAL IMPACT DASHBOARD

ONE-PAGE EXECUTIVE SUMMARY | Power BI • Data Analytics Project

Project Overview

This project uses a cleaned, country-and-date COVID-19 dataset to build an interactive Power BI dashboard for exploring reported cases, deaths, and vaccination progress across countries and over time. The dashboard is designed to help users identify trends, compare country-level impacts, and review vaccination coverage.

Objectives

- Track monthly and yearly changes in newly reported COVID-19 cases.
- Compare reported cases and deaths across countries and regions.
- Identify countries with high reported case and death totals.
- Explore vaccination progress using fully vaccinated people and coverage per 100 people.
- Enable focused analysis with date, country, and region slicers.

Dashboard Components

Visual	Purpose
KPI cards	Summarize cases, deaths, and vaccination indicators using appropriate latest-date values.
World map / country bar charts	Compare geographic distribution and rank countries by reported impact.
Yearly and monthly trend charts	Show changes in newly reported cases over time.
Cases vs. deaths chart	Explore how reported cases and deaths move across periods.
Vaccination coverage chart	Compare fully vaccinated people per 100 rather than relying only on population-sized totals.

Key Analytical Insights

- **Trend analysis:** Monthly and yearly views can reveal periods of increased or decreased reported cases; confirm peak periods from the finished visuals.
- **Geographic comparison:** Country rankings and maps highlight differences in reported impact, while population size and reporting practices affect comparisons.
- **Vaccination comparison:** Coverage per 100 people is generally more informative for cross-country comparison than raw vaccinated counts alone.

Data Quality & Interpretation

The cleaned dataset contains 58,560 case-reporting rows across 240 countries/areas, covering 3 January 2020 to 5 December 2024. The project checks date validity and duplicate country-date records. The prepared data previously showed 0 duplicate country-date pairs and no invalid dates. Some fields contain substantial missingness, especially vaccination indicators; missing values must not be interpreted as zero. Cumulative measures must be taken from an appropriate latest available record rather than summed across daily history, which would double-count.

Outcome: A single-page executive view that supports trend monitoring, country comparison, and vaccination analysis. Findings should be interpreted as reported data—not as a complete measure of actual infections or vaccine uptake.

Prepared for project documentation • Tool: Power BI • Source: cleaned COVID-19 cases and vaccination dataset. This summary describes the analysis design; chart-specific rankings and peak values should be populated from final dashboard results.